

What's the Difference? Reconciling Knowledge Structure Concepts to Aid AI Development for Human-Machine Teaming

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A significant challenge for the development of artificial social intelligence for effective human-machine teams is defining the forms of artificial knowledge structures needed for machine agents to meaningfully engage in collaboration. Relevant to this, individual and shared knowledge structure concepts have been proposed across a variety of disciplines, resulting in a lack of conceptual clarity and impeding their operationalization for human-machine teaming. To reconcile conceptual differences across disciplines and enable the emergence of complex socio-cognitive abilities in machine agents, research is needed to integrate theory on the knowledge structures that underpin complex cognition. Toward this end, we survey research from the cognitive and computational sciences to develop a framework for the systematic application and evaluation of knowledge structure concepts for machine agents in teams. Our approach focuses on contextual factors, specifically the task environment structure and the situation temporality, that can help guide knowledge structure requirements for artificial social intelligence.

Teams continue to be a primary mode of work for tackling society's complex problems. Teams are defined as a particular type of group (i.e., two or more individuals) characterized by interdependence, that interacts and adapts to perform their responsibilities and achieve shared goals (Sundstrom et al., 1990; Salas et al., 1992). Teams rely on intensive levels of communication coming from multiple information sources. Here, they draw from member task-relevant knowledge to support the interdependencies between their taskwork. To be effective, though, they must coordinate so as to integrate their specialized roles towards their common aim.

Teamwork increasingly relies on existing and emerging technologies to support achieving objectives (Fiore et al., 2010; Salas et al., 2012). Such technologies can support traditional collocated collaborations but also enable distributed and asynchronous collaborations, critically scaffolding the knowledge building activities of teams. At the same time, strides made in the development of artificial intelligence have led to the increasing integration of machine agents in teams to support human work (Fiore & Wiltshire, 2016). This requires improved understanding of artificial social intelligence (ASI), as well as selecting and defining the forms of artificial cognition pertinent to collaborative processes.

In research on human teams, complex social cognitive abilities have been linked to both team performance (Woolley et al., 2015) and team viability (Whiting et al., 2019). This work shows that teams composed of individuals with higher social intelligence tend to exhibit higher levels of collective intelligence, defined as "the general ability of a particular group to perform well across a wide range of different tasks" (Woolley et al., 2015, p. 420). Social intelligence, in this research context, is the ability of individuals to accurately perceive and identify the mental states of other people on the basis of emotional displays.

In human-machine teaming research, there has been a significant effort made to transition machines from tools to

facilitators of collaborations to, eventually, teammates in their own right (Phillips et al., 2011; Warta et al., 2016). For machines to be effective collaborators, the instantiation of both social and cognitive competence is paramount (Wagner & Briscoe, 2017). A key gap is defining the forms of knowledge that must be developed and maintained in support of ASI. To address this gap, it is necessary to produce a set of requirements that guide the development of knowledge structures in machine agents. For complex social cognition in particular, machine agents must be endowed with the capability to develop and draw on knowledge structures relating to their individual teammates and the interactions between them, as well as the task environment. We contend that, to facilitate the emergence of complex social cognitive abilities in machine agents, research is needed to integrate theory on the knowledge structures that underpin complex cognition and reconcile conceptual differences across disciplines. The purpose of this paper is to provide guidance on the facets of individual and team cognition that can be targeted for ASI. To this end, we describe how differences in task environment, and the temporal nature of the situation, can help elucidate distinct forms of knowledge structures that can support developing ASI for human-machine hybrid teams.

KNOWLEDGE STRUCTURE FOUNDATIONS

In this section, we survey and integrate theory from several disciplines examining foundational cognitive processes to assess differences in terminology and definitions developed to describe the structural knowledge used to guide behavior. This survey focuses on forms of knowledge used to describe conceptual structures that humans rely on to store and apply information; namely, schemas, scripts, and mental models. Although these vary to some extent, these knowledge structures help people: (1) organize their understanding of the world; (2) make interpretations of events and predictions of future states; and (3) decide how to act in particular contexts.

Schemas

Schema theory has its origins in cognitive psychology, with one of the most widely-cited definitions of schema provided by Mandler (1978): "a spatially and/or temporally organized structure in which the parts are connected on the basis of contiguities that have been experienced in space or time" (p. 263). Across definitional variants proposed in the literature, schemas are conceptualized as being, in some sense, fixed in space and time although the experience represented is recurrent. The subcomponents of a schema are the particular features that describe an experience, enabling the development of expectations with respect to that experience and how it is likely to unfold (Frederick, 1991), even if all such features are not made explicit. These features are identified and selected based on memory of previous instances of the event.

Scripts. In seminal work on knowledge structures in humans and artificial intelligence, Abelson (1981) described at length one particular type of schema, the script—"a cognitive structure that, when activated, organizes comprehension of event-based situations" (p. 717). This notion of a script emphasizes the ordering of actions toward a goal (i.e., a causal chain of events). These can be viewed as a list of actions a person follows to achieve a desired goal and they capture general information about a routine series of events or recurrent type of social event. For example, a restaurant script consists of "entering, ordering, eating, paying, and leaving". A challenge for modeling is that scripts contain particular sorts of information (Schank & Abelson, 1977). This includes identifying a name or theme, with typical roles and entry conditions. There is also a sequence of goal-directed scenes and sequences of actions within each scene. Here, each sequence of actions has the property of a causal chain in that every preceding action provides conditions for performing consequent actions.

More recently, schemas and scripts have been implemented in, for example, deep learning approaches attempting to flexibly apply preconstructed knowledge structures (Kim et al., 2016). Other lines of work argue for a need to develop "computational narrative intelligence" (i.e., a machine's ability to comprehend a story; Riedl, 2016). However, like both collective intelligence and ASI, this necessitates social sensitivity at the level of the individual agent, human or machine.

Mental Models

Much of the foundational work on mental models spans across major subdisciplines within psychology—cognitive, organizational, and human factors—and the phrase mental model grew out of early work in cognitive science, with credit for the phrase typically granted to Craik (1943), and brought to prominence decades later in cognitive psychology (e.g., Johnson-Laird, 1983, 1989) and human factors psychology (e.g., Rouse & Morris, 1986). In cognitive psychology, mental models have been implicated in the knowledge integration processes that afford comprehension of a phenomenon and, in turn, the ability to draw inferences (Johnson-Laird, 1983) and

make predictions within some context (Getner, 2002). In the organizational psychology literature, mental models have been characterized in terms of the environment, where components of the environment and their relationship to each other are represented in the knowledge structure (Klimoski & Mohammed, 1994; Mohammed & Dumville, 2001). Lastly, in human factors psychology, descriptions of mental models are largely functional, emphasizing the actions afforded by internal representations of a system (i.e., generating descriptions, explanations, and predictions) created by integrating knowledge derived from prior experience (i.e., activation in long-term memory) and based on observations of recent past and ongoing events.

In research on human-robot interaction, mental model components have been represented in machine agents using propositions and probabilistic statements. The implementation of these models though is challenging, in part due to a lack of conceptual clarity (Wagner & Briscoe, 2017). Furthermore, attempts to develop mental models in machine agents tend to be task-oriented (e.g., Goodrich & Yi, 2013).

PROPOSED FRAMEWORK

In this section, we offer a notional reconciliation of the aforementioned concepts to foster productive interdisciplinary research on ASI. Although computer scientists have implemented various forms of schemas, scripts, and mental models, they have not reconciled the theoretical differences and applications of these concepts for ASI. What is additionally needed, is attention to foundational concepts in team cognition theory. In this line of research, individual-level knowledge structure concepts have been adapted to explain and improve shared cognition for effective collaborations. Accordingly, teams are characterized by several key features: members have roles necessitating specialized knowledge and skills that are complementary to their teammates; teammates interact dynamically towards a common goal; and teammates' work requires shared and/or complementary mental models.

We build on prior theorizing to explicate the development and use of shared and distributed knowledge structures for human-machine teaming. Here, one can distinguish between the degree to which there is overlap between teammates' knowledge structures and the degree to which there is complementarity between teammates' knowledge structures. In the case of overlap, researchers refer to *shared* mental models and assess the *sharedness* of teammates' mental models. In the case of complementarity, researchers refer to *complementary* mental models as *transactive* memory systems. These higher-order knowledge structures guide teammates' assumptions, expectations, and behavior regarding the process of working as a team. We next provide a theoretical framework that can be used to parse the conceptual space associated with types of knowledge structures. Following this, we show how our framework can be integrated with team cognition research such that it can help describe the form of shared knowledge required by agents in human-machine teams.

Clarifying Conceptual Spaces

We propose a framework grounded in contextual factors to systematically evaluate the application of concepts describing the knowledge structures that underpin comprehension and behavior. The purpose of this framework is to move the field forward through a more seamless integration of the cognitive and computational sciences for the purposes of developing ASI. We contend that the characteristics of both the task and situation at hand dictate the form of knowledge structure that is drawn on to enable understanding and, when appropriate, guide subsequent action.

First, we address why there are differences across literatures in the use of these concepts. Specifically, differences in the origin and use of concepts may be due to the kinds of tasks that have been studied in varied disciplines. When examining the research contexts in which schemas, scripts, and mental models were used, there are consistent differences across settings. Schema and script concepts were primarily used in research where the environment was more well-structured (e.g., a restaurant script). The mental model concept was primarily used in more ill-structured environments (Rouse & Morris, 1986). For example, Brewer (1987) argued that a mental model could be seen as the activation in working memory of task-related long-term memory in a dynamic context (e.g., operating a control panel).

Second, with that theoretical distinction as our stepping off point, we can use this to juxtapose and define differently structured contexts. When an environment is well-structured, one is more easily able to retrieve and act on prior knowledge; that is, prior knowledge can be retrieved intact. But, in ill-structured environments, one needs to engage in a contextualized construction of knowledge. Although aspects of long-term memory are activated, these are those most immediately relevant to the situation, which still need to be adapted given the present ambiguity. The structure of the task thus determines knowledge requirements. In well-structured environments, one can draw from inert and readily accessible forms of knowledge whereas, in ill-structured environments, one has to rely on dynamically constructed and more ephemeral contextualized knowledge.

Third, we can further consider the temporal demands of the environment, that is, whether it is a relatively static or dynamic situation. In some situations, the expected sequence of events might be fixed and the actions available to the agent may be limited based on a number of factors, including feasibility and defined goals. In these static situations, the agent can rely on their inert knowledge, drawing out relevant facts and previously acquired information from long-term memory to formulate sufficient comprehension and make a decision. In other situations, there might be some ambiguity concerning the precise or correct sequence of events, perhaps as a result of missing information or the inherent complexity of the environment such that the set of potential actions available to the agent is relatively unbounded beyond achieving a predefined goal. Here, the agent must recognize how differing cues and environmental input differentially

align with their prior knowledge and what new knowledge must be acquired.

Integrating Conceptual Spaces

We next integrate these distinctions as dimensions in a framework that clarifies when the concepts of schema and mental model could be more clearly applied. As illustrated in Table 1, these dimensions can be used to create a 2x2 conceptual space juxtaposing well-structured versus ill-structured environments and static versus dynamic situations. Specifically, this framework focuses on the *structure* of the task environment along with the *temporal nature* of the situation. Given that schemas are generally conceptualized as fixed yet recurrent, the application of this knowledge structure concept may be better suited to understand the means through which agents achieve comprehension of situations that are familiar and have minimal, if any, uncertainty (i.e., well-structured and static). In contrast, mental models are more appropriately applied in ill-structured environments in that they are conceptualized as developing over time with continued updates to store and understand new information via activation in long-term memory as it is received by the agent. Indeed, with this form of knowledge structure, an agent could develop, for example, cue-strategy associations that may enable predictions and ultimately guide the selection of appropriate action.

Table 1. Proposed framework for human-machine teaming research on specification of knowledge structure requirements.

		Nature of Situation	
		Static	Dynamic
Task Environment	Well-structured	Schema	
	Ill-structured		Mental Model

We also suggest that these conceptual differences can be adapted for team research to provide a foundation for developing ASI. Specifically, we draw from Fiore et al. (2003), who argued that a shared mental model (SMM) can be viewed as the dynamic activation in working memory of both team-relevant and task-related knowledge while engaged in an ill-structured context. In this view, the SMM is constructed dynamically based on one’s current situation and interpretation of task needs in conjunction with long-term memory. Importantly, overlap between the dynamic mental model of each team member facilitates coordination and effective collaborative behavior towards a shared goal (Fiore et al., 2003). This can be contrasted with a team schema, a more inert form of knowledge about teammates’ expertise and capabilities that can be relied upon in more well-structured contexts. In this way, it is more like transactive memory systems, a form of long-term memory capturing the complementary knowledge of team members, which supports the coordination of expertise.

With this as a foundation, we offer the following research questions and notional requirements to guide the development of ASI at the individual and team level. This, then, can be leveraged in support of research and development for socially-intelligent machines. Note that the diagonals in Table 1 are blank because research is needed to explicate the applicability of schemas or mental models in these cells. As such, these research questions represent an initial map of the problem space that can integrate individual and team cognitive theory where long-term and working memory are utilized in support of teamwork in complex operational environments.

- **RQ 1.** How can schema theory and scripts inform the specification of knowledge structure requirements for human-machine teaming in *routine* tasks and situations (i.e., conditions are relatively static and well-structured)?
 - RQ 1.1. What are the schema requirements for machine agent *comprehension* of a *routine* situation?
 - RQ 1.2. What are the schema requirements for machine agent *response* to a *routine* situation?
- 1. *Notional Schema Requirements for Machine Teammates*
 - 1.1. Scripts that underpin understanding of recurrent tasks. Representation of relationship between a set of events that facilitates comprehension of causal dependence and identification of next event given the sequence of previous events.
 - 1.2. Schemas that underpin understanding of established social and task roles. Representation of relationship between teammates and their tasks that facilitates identification of pooled and sequential coordination needs (i.e., understanding roles and tasks with low levels of interdependence).
 - 1.3. Scripts that underpin understanding of recurrent team interactions. Representation of relationship between team member actions and observed outcomes that facilitates identification of goal-appropriate behavior, both in terms of team processes (e.g., communication) and task outcomes (e.g., performance).
- **RQ 2.** How can mental model theory and schema theory be integrated to inform the specification of requirements for human-machine teaming in *novel* tasks and situations (i.e., conditions are relatively dynamic and ill-structured)?
 - RQ 2.1. What are the knowledge structure requirements for machine agent *comprehension* of a *novel* situation?
 - RQ 2.2. What are the knowledge structure requirements for machine agent *response* to a *novel* situation?
- 2. *Notional Mental Model Requirements for Machine Teammates*
 - 2.1. Model of teammates (individuals). Activation of working memory based on observations of cue/environmental change(s) and activation of role and team interaction long-term memory supports formulation of cue-strategy associations. This could be used to help the agent understand, predict, and

respond to, for example, irregular teammate behavior, team newcomers' traits and dispositions, and roles that emerge over time.

- 2.2. Model of teammate interactions (relationships). Activation of working memory based on observations of cue/environmental change(s) and activation of routine task and team interaction long-term memory supports formulation of cue-strategy associations. This could be used to help the agent understand, predict, and respond to, for example, a team's complex coordination needs (high levels of task interdependence), or emerging norms.
- **RQ 3.** How can theoretical and empirical work on shared and complementary knowledge structures inform the specification of requirements for human-machine teams in *novel* tasks and situations?
 - RQ 3.1. How are shared versus complementary knowledge structures retrieved and constructed for human-machine teams' *comprehension* of a *novel* situation?
 - RQ 3.2. How do shared versus complementary knowledge structures support human-machine teams' *response* to a *novel* situation?
3. *Notional Shared Knowledge Structure Requirements for Machine Teammates*
 - 3.1. Model of specialized and complementary knowledge distribution. Complementary knowledge structures may better support comprehension; machine agents should track and prompt information exchange related to specializations to support teamwork (e.g., use transactive memory search to accelerate knowledge building processes).
 - 3.2. Model of knowledge redundancy. Shared knowledge structures may better support action; machine agents should track overlap and prompt knowledge building to support teamwork (e.g., to reach consensus, monitor cohesion, etc.).

CONCLUSION

Technological advances have fundamentally altered the nature of work and continue to do so, both necessitating and enabling novel human-machine configurations. In their current state, intelligent machine agents have been introduced in collaborative work domains as a means to improve taskwork by, for example, minimizing human workload through the automation of routine tasks (Lebeuf et al., 2018). However, such agents are also perceived as lacking sufficient intelligence for decision-making tasks and their social errors only serve to highlight a larger gap in socio-cognitive abilities that enable collaboration (e.g., Wessel et al., 2018). We contend that differences in the definition and application of knowledge structure concepts can be leveraged to guide the development of ASI. Based on this, we drew on contextual dimensions to propose a framework for the systematic evaluation and application of knowledge structure concepts for socially-intelligent machine agents. We then illustrated how

this framework can guide future research by outlining a set of questions and notional knowledge structure requirements for human-machine teams.

Our approach fits with recent calls for explanatory pluralism to capture the complexity inherent in social cognition (Wiltshire, Lobato, McConnell, & Fiore, 2015; Wiltshire, Warta, Barber, & Fiore, 2017). Humans likely employ different processes depending upon the social cognitive functions needed. Research is needed to integrate these mechanisms and empirically demonstrate the contexts and situations in which one particular knowledge structure would be adopted as opposed to another. This is critical to the development of ASI in that architectures need to be flexible enough to not only learn when different mechanisms are applied, but also understand the nuanced behavioral characteristics to be monitored when collaborating with human team members.

In sum, to advance artificial social cognition, not only is a closer collaboration between the social, cognitive, and computer sciences needed, but that collaboration must build off a deep understanding of the rich literature examining cognition and collaboration. Our framework represents one such approach and helps guide research drawn from the debates about the basic fundamental mechanisms of social cognition. The addition of basic long-term memory knowledge structures related to social-cognitive knowledge could increase collaborative competence of non-human members of human-machine teams. There is a deep base of theoretical and empirical research in cognition and collaboration, with various nuanced associations and commitments aligned with differing approaches. We have summarized key knowledge structure differences in socio-cognitive mechanisms to provide conceptual grounding for the research on which to draw to both guide the development of notional socio-cognitive architectures and generate research questions about social cognition in artificial systems.

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